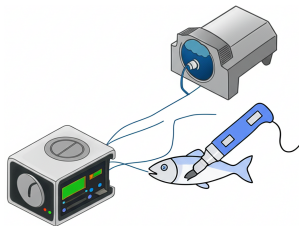


# Deep Learning for Marine Biomass Analysis

Using Rapid Evaporative Ionization Mass Spectrometry



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Bing Xue  
Mengjie Zhang  
Daniel Killeen

# Batch Detection for Marine Biomass

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# Batch Detection for Marine Biomass

Batch Detection  
for Marine  
Biomass

The SpectroSim  
Architecture

Our Model  
Achieves 98%  
Accuracy

Conclusion

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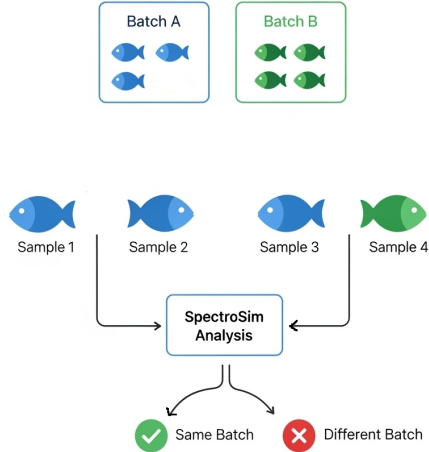
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## Batch Detection for Marine Biomass



- ◇ **The Problem:** Detecting the specific processing batch of a fish is critical for food safety and fraud prevention, but it's an unaddressed research problem.
- ◇ **Data Challenges:** Rapid Evaporative Ionization Mass Spectrometry (REIMS) data is:

**High-Dimensional:** 2,080 features from only 72 initial fish samples.

**Noisy and Sequential:** Traditional machine learning methods struggle to capture the complex patterns.

- ◇ **Our Proposal:** We introduce SpectroSim, a novel self-supervised contrastive learning framework to determine if two fish samples come from the same batch.

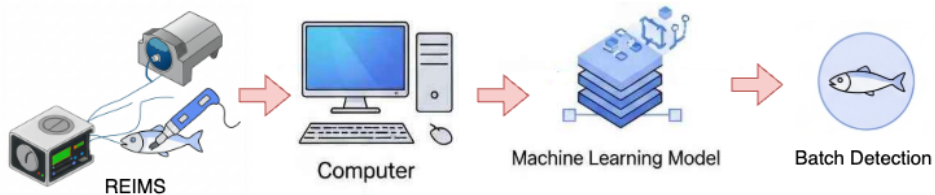
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**Figure 2.** REIMS-based batch detection

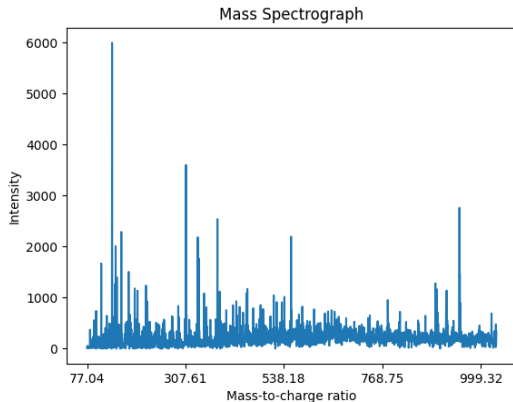
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**Figure 3.** Rapid Evaporative Ionization Mass Spectrometry (REIMS)

# The SpectroSim Architecture

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- ◇ **Core Idea:** SpectroSim learns to create similar embeddings for fish from the same batch and different embeddings for fish from different batches, without using any class labels during training.
- ◇ Key Innovations:

**Transformer Encoder:** Instead of a standard image-based (ResNet) backbone, we use a Transformer architecture to specifically capture the long-range, sequential patterns in the 1D mass spectrometry data.

**Custom Projection Head:** The projection head is custom-designed for mass spectrometry data to reduce information loss.

**Efficient:** The model works effectively with a small batch size of 16, making it computationally accessible.

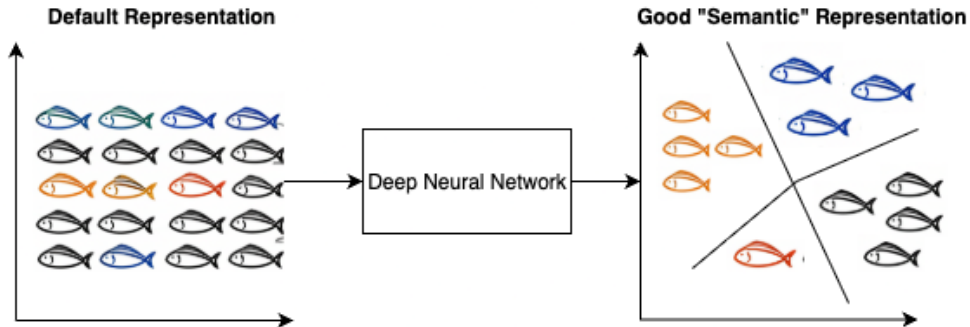
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Biomass

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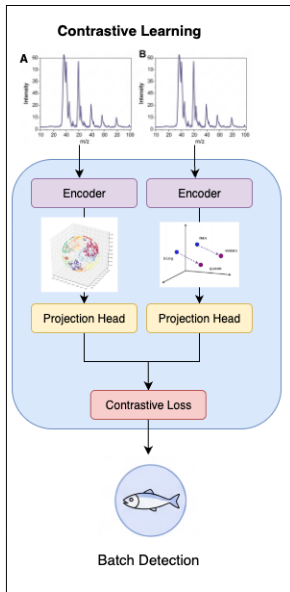
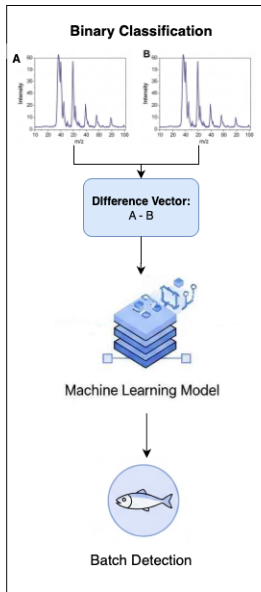
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The SpectroSim  
Architecture

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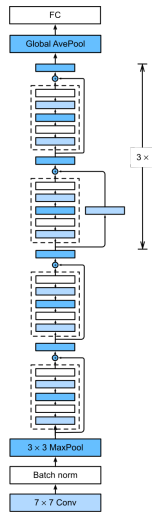
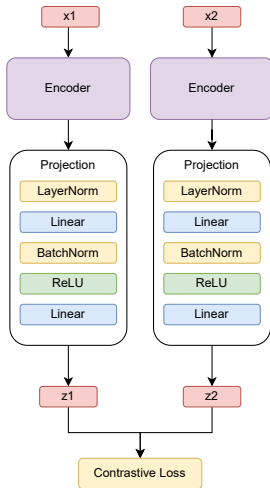
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Biomass

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Architecture

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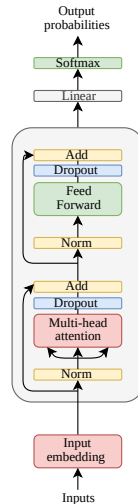
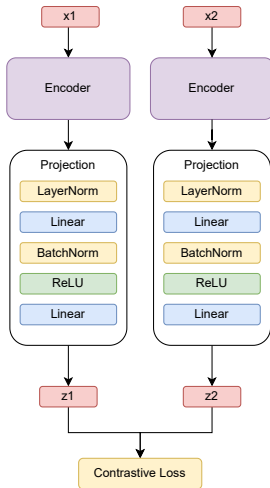
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Biomass

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# Out Model Achieved 98% Accuracy

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| Method            | Binary Classification                |                                     | Contrastive Learning                 |                                     |
|-------------------|--------------------------------------|-------------------------------------|--------------------------------------|-------------------------------------|
|                   | Train                                | Test                                | Train                                | Test                                |
| OPLS-DA           | 57.08% $\pm$ 1.46                    | 53.19% $\pm$ 2.22                   |                                      |                                     |
| KNN               | 100.00% $\pm$ 0.00                   | 55.69% $\pm$ 2.74                   |                                      |                                     |
| DT                | 100.00% $\pm$ 0.00                   | 51.77% $\pm$ 1.43                   |                                      |                                     |
| LDA               | 89.13% $\pm$ 1.15                    | 56.35% $\pm$ 2.70                   |                                      |                                     |
| NB                | 66.95% $\pm$ 2.54%                   | 55.01% $\pm$ 3.26%                  |                                      |                                     |
| RF                | 99.87% $\pm$ 0.33                    | 50.02% $\pm$ 0.14                   |                                      |                                     |
| SVM               | 92.44% $\pm$ 1.05                    | 54.19% $\pm$ 2.87                   |                                      |                                     |
| LR                | 91.41% $\pm$ 1.19                    | 53.90% $\pm$ 3.12                   |                                      |                                     |
| Ensemble          | 95.05% $\pm$ 0.99                    | 54.14% $\pm$ 2.81                   |                                      |                                     |
| CNN               | 96.12% $\pm$ 5.48                    | 58.09% $\pm$ 1.01                   | 100.00% $\pm$ 0.00                   | 96.87% $\pm$ 4.30                   |
| R-CNN             | 100.00% $\pm$ 0.00                   | 51.01% $\pm$ 1.39                   | 100.00% $\pm$ 0.00                   | 84.98% $\pm$ 3.37                   |
| KAN               | 100.00% $\pm$ 0.00                   | 89.49% $\pm$ 2.50                   | 100.00% $\pm$ 0.00                   | 93.75% $\pm$ 3.24                   |
| LSTM              | 100.00% $\pm$ 0.00                   | 53.03% $\pm$ 0.90                   | 100.00% $\pm$ 0.00                   | 96.87% $\pm$ 5.63                   |
| Mamba             | 100.00% $\pm$ 0.00                   | 56.26% $\pm$ 2.25                   | 100.00% $\pm$ 0.00                   | 93.75% $\pm$ 4.11                   |
| VAE               | 100.00% $\pm$ 0.00                   | 50.51% $\pm$ 1.02                   | 100.00% $\pm$ 0.00                   | 67.83% $\pm$ 3.01                   |
| SimCLR            |                                      |                                     | 100.00% $\pm$ 0.00                   | 84.98% $\pm$ 3.37                   |
| <b>SpectroSim</b> | <b>100.00% <math>\pm</math> 0.00</b> | <b>95.77% <math>\pm</math> 2.22</b> | <b>100.00% <math>\pm</math> 0.00</b> | <b>98.02% <math>\pm</math> 1.71</b> |

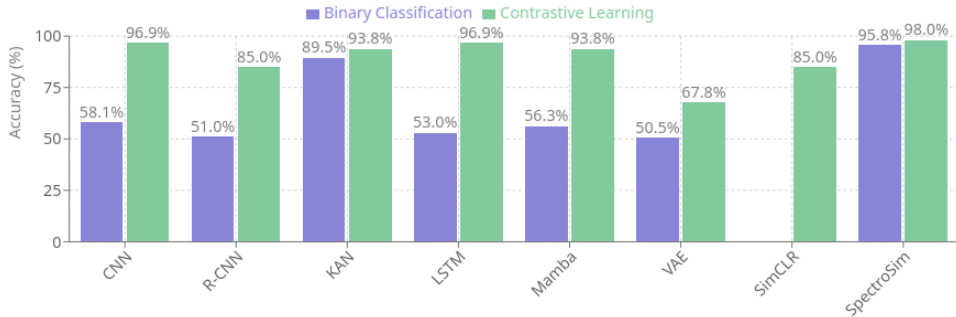
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**Figure 5.** Model Performance: Binary Classification vs. Contrastive Learning



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- ◇ **Traditional Methods Fail:** Standard models like KNN and Decision Trees overfit severely, achieving only 50-56% test accuracy.
- ◇ **SpectroSim Excels:** In a self-supervised contrastive learning setup, SpectroSim achieves 98.02% test accuracy.
- ◇ **Contrastive Learning is Superior:** Across all deep learning models tested, switching from simple binary classification to contrastive learning dramatically improved performance.

For example, an R-CNN model jumped from 51.01% to 84.98% accuracy.

The Transformer itself improved from 95.77% (binary) to 98.02% (contrastive).

# Conclusion

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- ◇ **Conclusion:** SpectroSim's Transformer-based, self-supervised approach is highly effective for batch detection using complex REIMS data, significantly outperforming previous methods.
- ◇ **Impact:** This demonstrates that deep, sequence-aware models are superior for this type of chemical analysis.
- ◇ **Acknowledgements:** This work was supported by the MBIE Fund and we thank Plant and Food Research for their collaboration.

**Thanks for your attention!**

Backup Slide